

Real-World Traffic: How Apps Behave on Campus vs. Home Wi-Fi

EC 441 Final Project Demo
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Capturing the Data

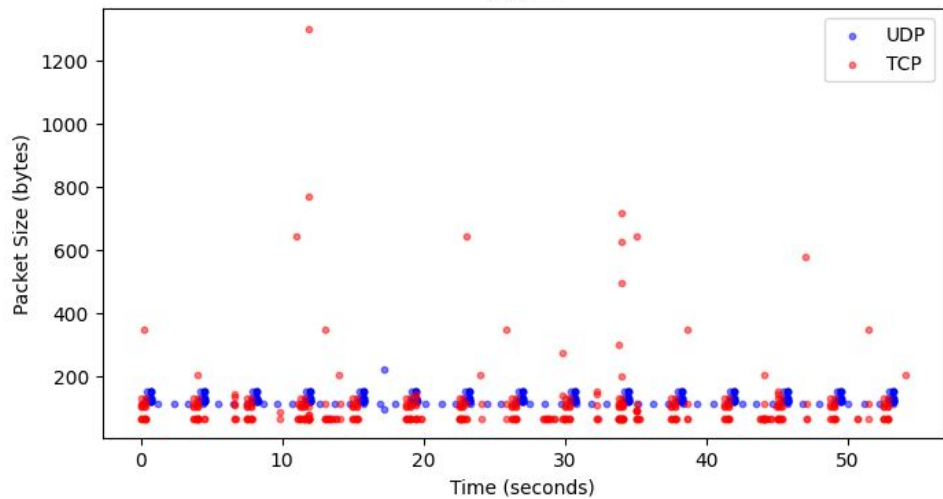
- Data Collection: Recorded live traffic using Wireshark to compare a high-density campus network against a residential Xfinity network.
- Test Scenarios: Captured four distinct usage profiles: Zoom, YouTube (Steady), YouTube (Surfing), and BBC News.
- Analysis Tools: Developed Python scripts leveraging pyshark and matplotlib for offline packet parsing and visualization.

Boston University Wi-Fi Network (eduroam)

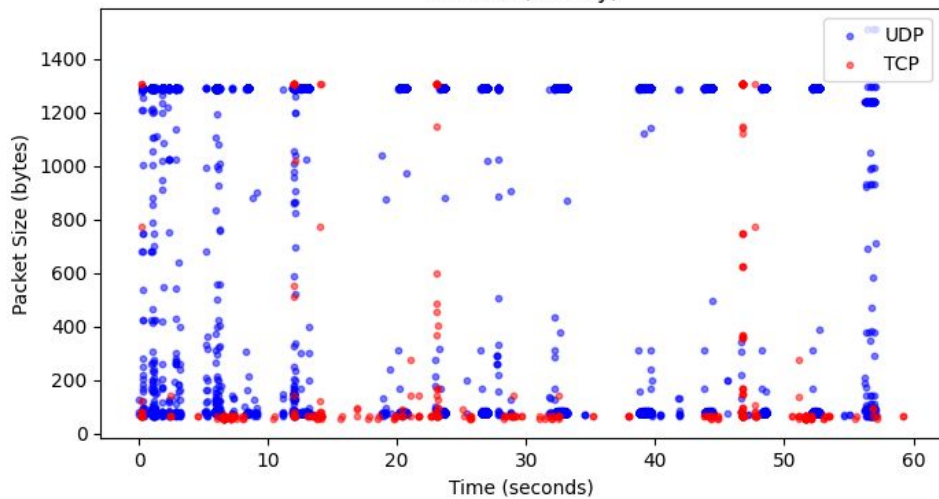
TCP vs. UDP

- Zoom (UDP): Prioritizes fast, real-time delivery by firing small, constant packets and ignoring drops.
- Web/Video (TCP): Ensures guaranteed reliability by sending massive data chunks for perfect reconstruction.
- Traffic Profiles: Web browsing generates highly bursty traffic, whereas live video requires a steady stream.

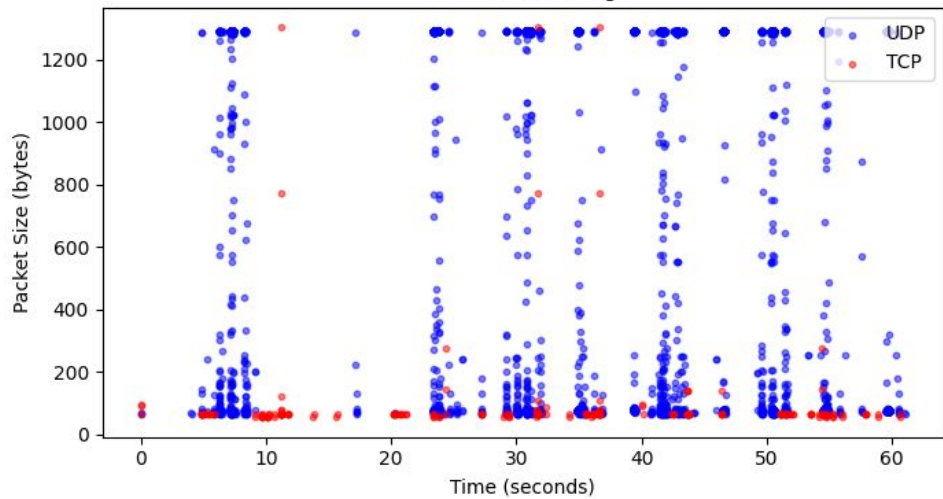
Zoom



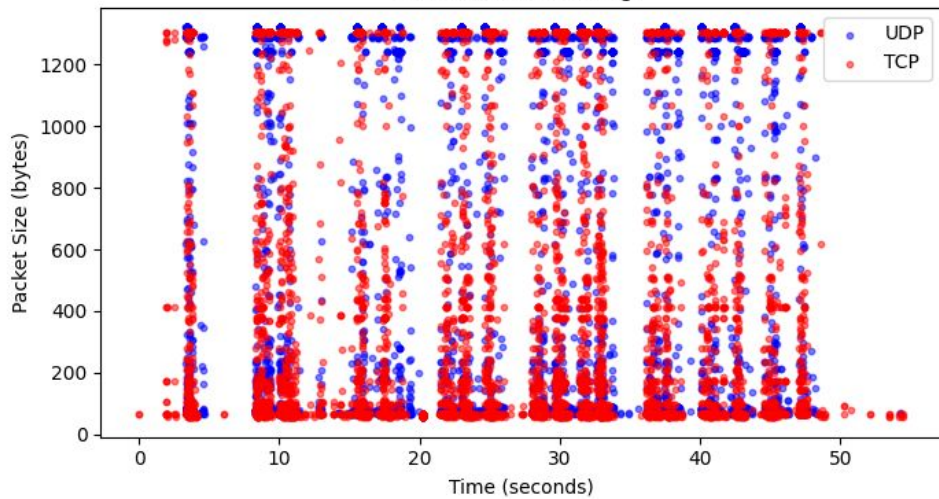
YouTube (Steady)



YouTube (Clicking)



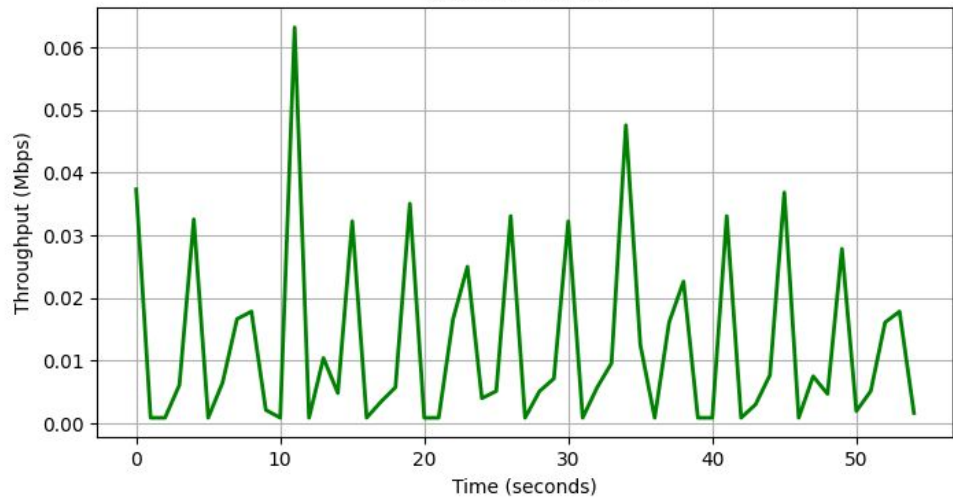
BBC News (Browsing)



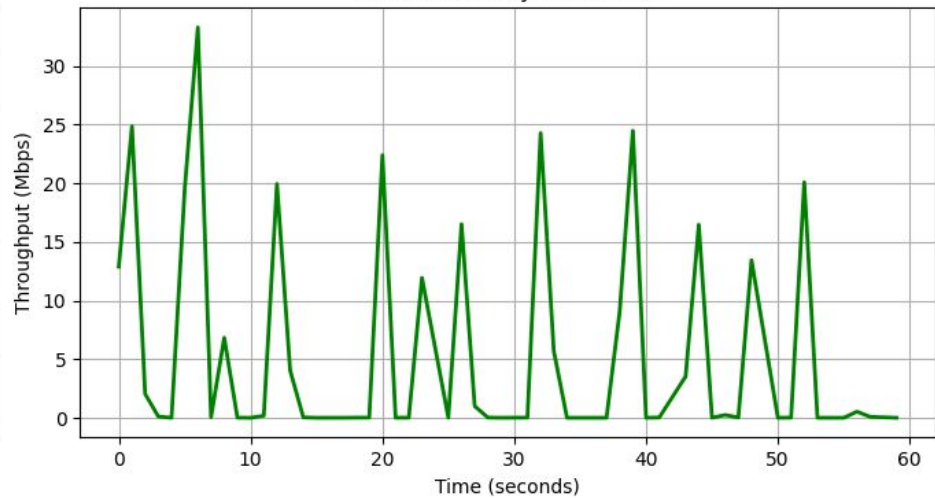
Bandwidth Comparison

- Zoom: Requires low but constant bandwidth (~1-3 Mbps) without buffering ahead.
- YouTube: Pulls massive data bursts (20+ Mbps) to fill the video buffer, then enters an idle state.
- BBC News: Exhibits distinct throughput spikes that correspond directly to user clicks and new page loads.

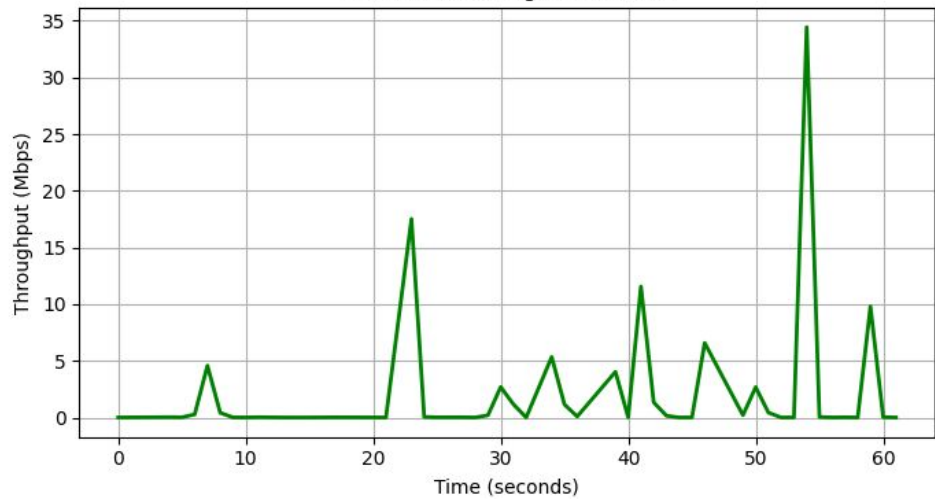
Zoom Bandwidth



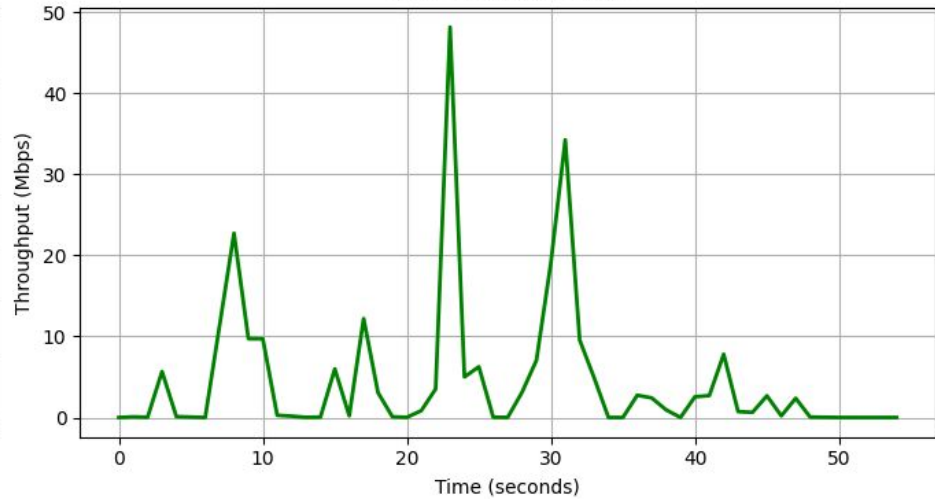
YouTube (Steady) Bandwidth



YouTube (Clicking) Bandwidth



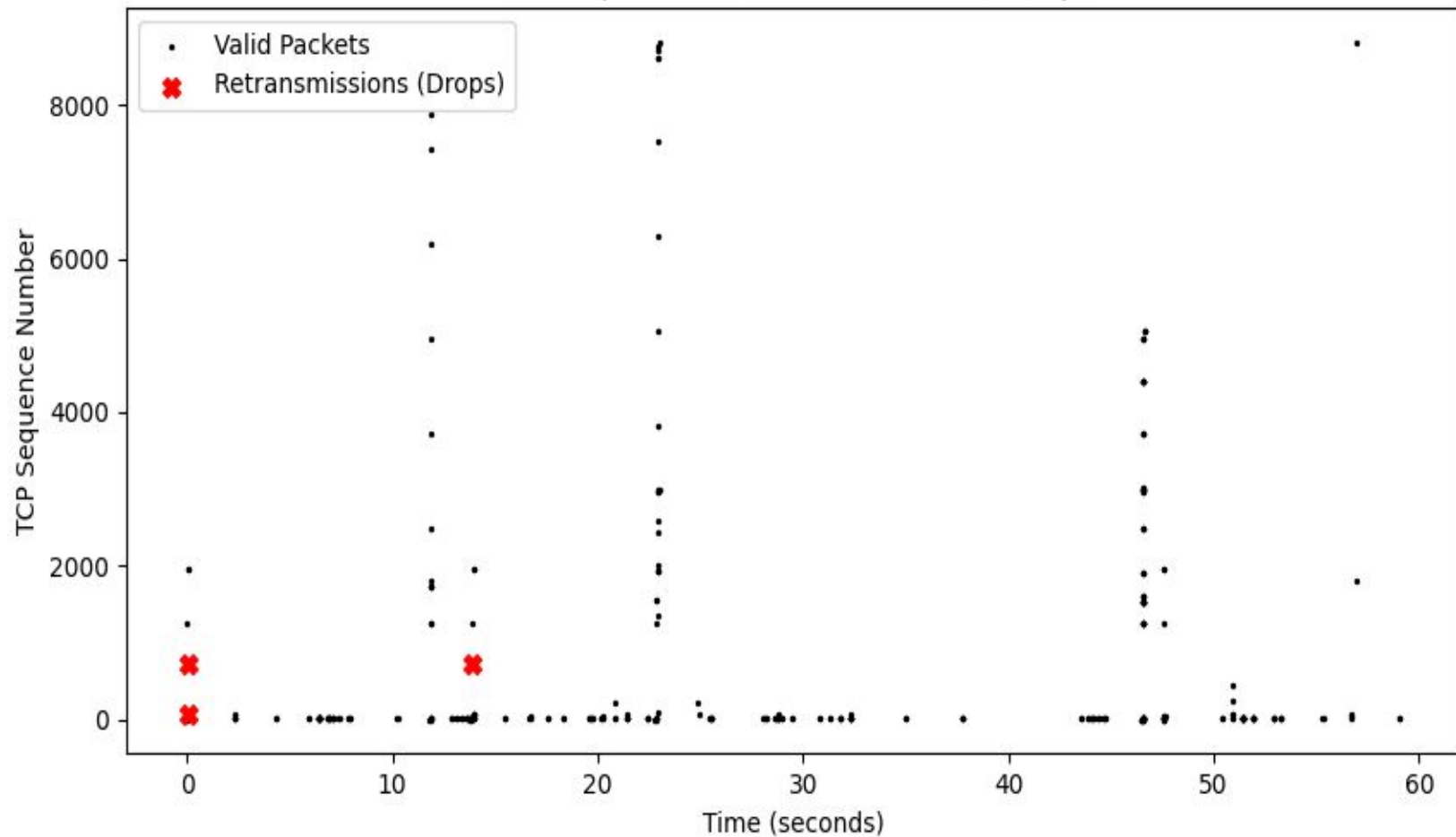
BBC News Bandwidth



TCP Packet Drops

- Transfer Trajectory: The rising black line indicates successful data transfer during video playback.
- Packet Loss: Red markers pinpoint exact timestamps where the campus network dropped a packet.
- Congestion Control: Every drop forces TCP to reduce its transmission window, throttling download speeds to protect network stability.

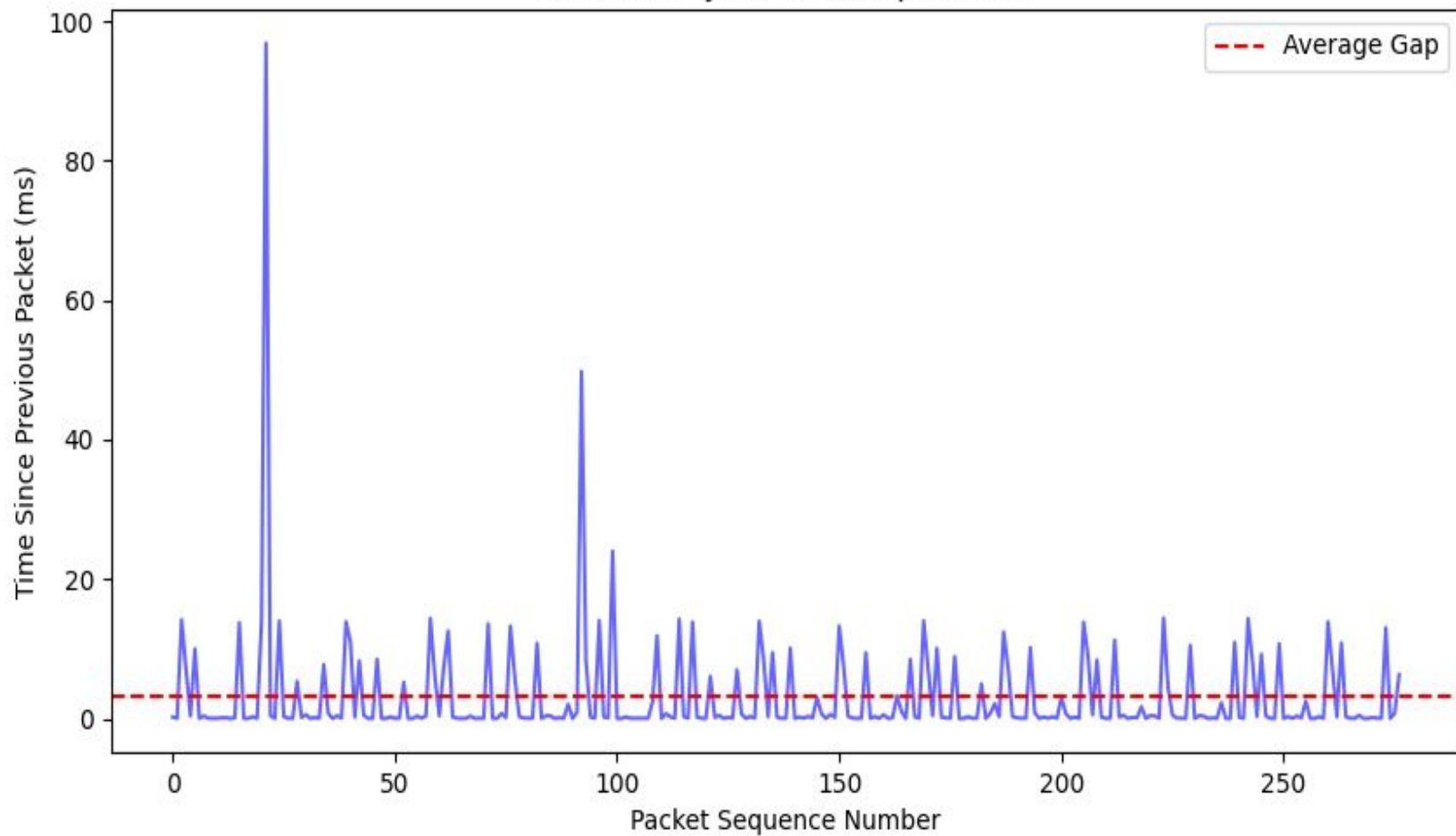
TCP Sequence Numbers and Packet Drops



UDP Jitter

- Timing Sensitivity: UDP skips retransmissions but relies heavily on precise packet timing.
- Interference Effects: Campus Wi-Fi congestion causes packets to arrive late or clump together, resulting in high jitter.
- User Experience: This erratic timing variance directly causes video freezing and audio stuttering during live calls.

Zoom UDP Jitter on Campus Wi-Fi

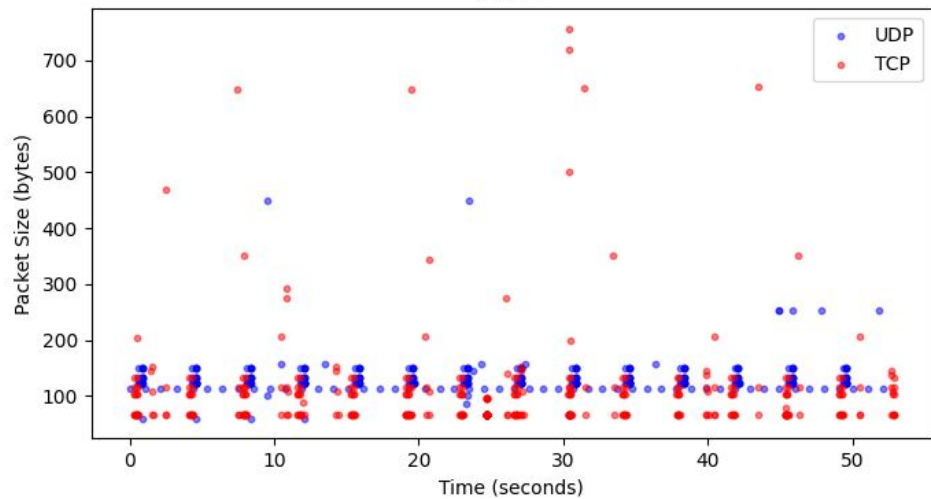


Home Wi-Fi Network (Xfinity)

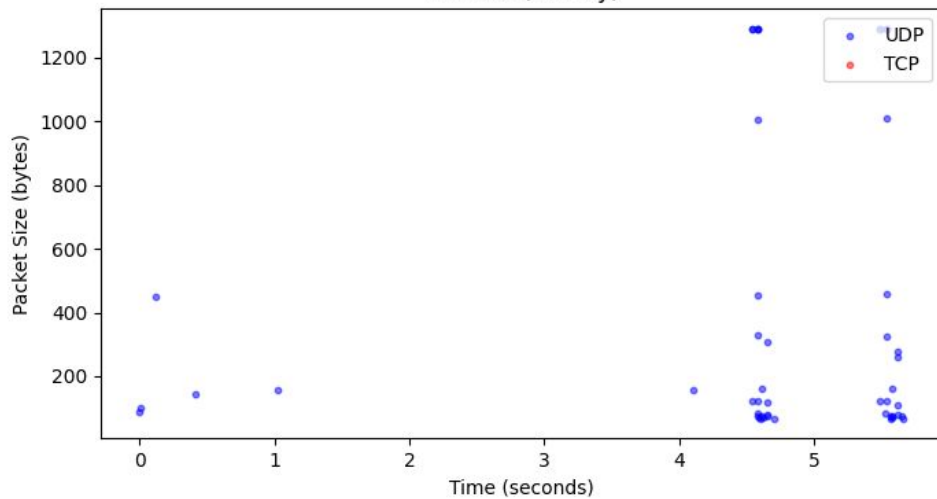
TCP vs. UDP

- Zoom (UDP): Maintains a consistent UDP profile, avoiding fallback mechanisms absent campus firewall interference.
- YouTube (TCP): Demonstrates clumping, downloading massive chunks before resting for several seconds.
- Clean Capture: Reveals significantly fewer stray packets from background services compared to the cluttered campus environment.

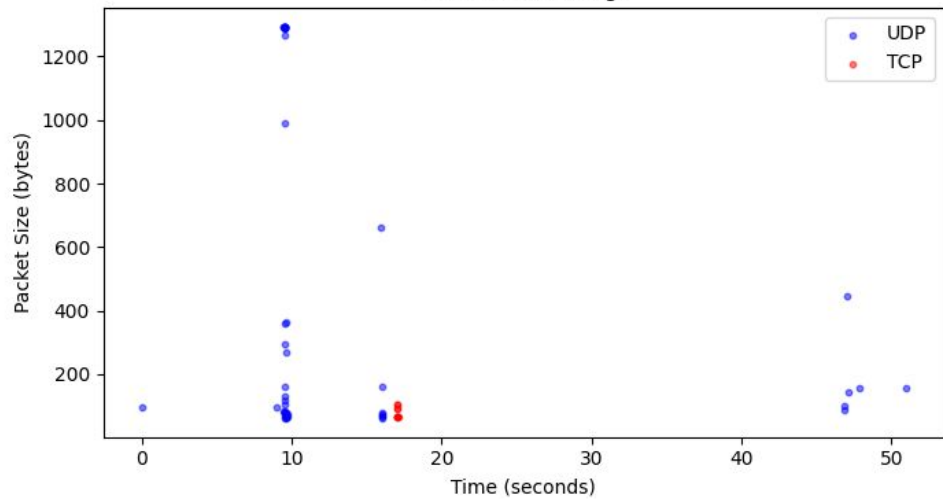
Zoom



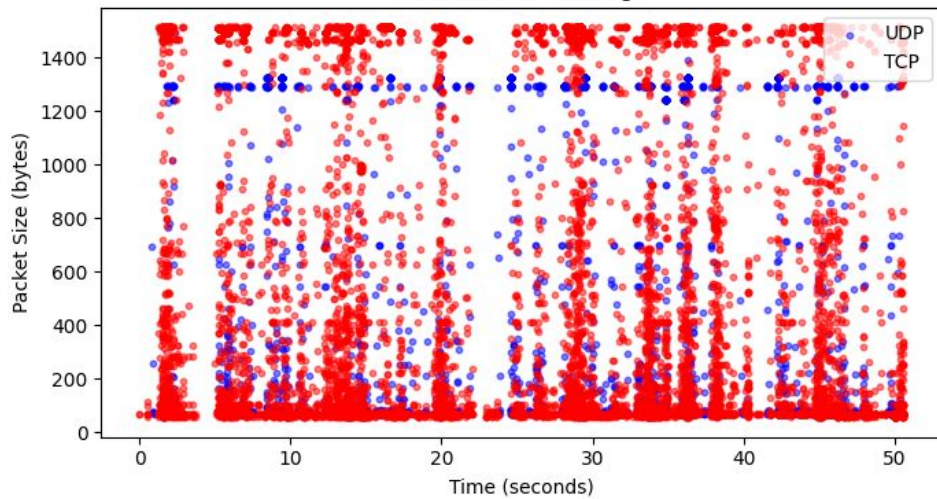
YouTube (Steady)



YouTube (Clicking)



BBC News (Browsing)

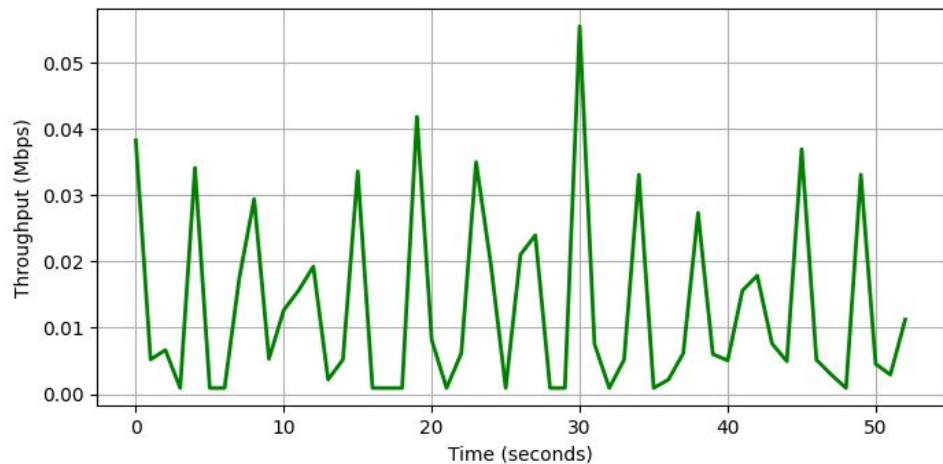


Bandwidth Comparison

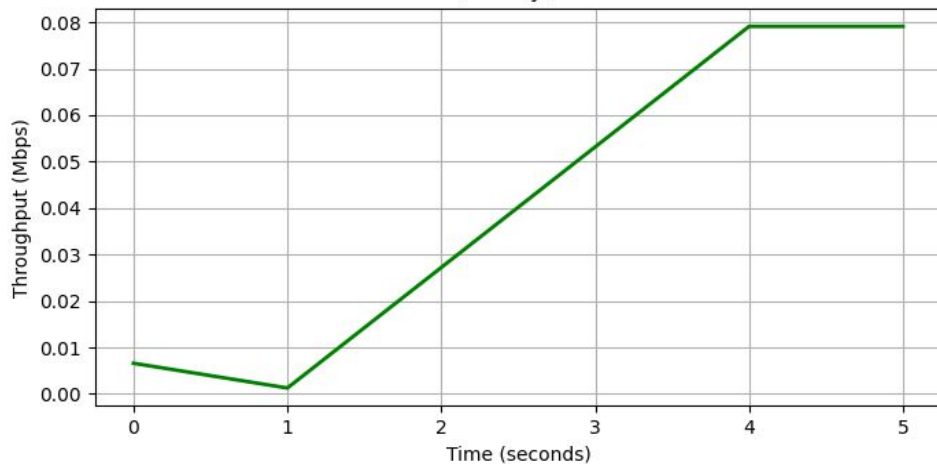
- YouTube Peaks: Reaches massive bursts of 40+ Mbps, nearly doubling the peak throughput observed on the campus network.
- TCP Efficiency: Throughput exhibits sharp, unhindered spikes, indicating the congestion window expands faster without airtime competition.
- Zoom Steady-State: Remains consistent at low throughput (~0.1 Mbps), confirming demand is dictated by the application codec rather than network capacity.

Application Bandwidth Demand (Home Wi-Fi)

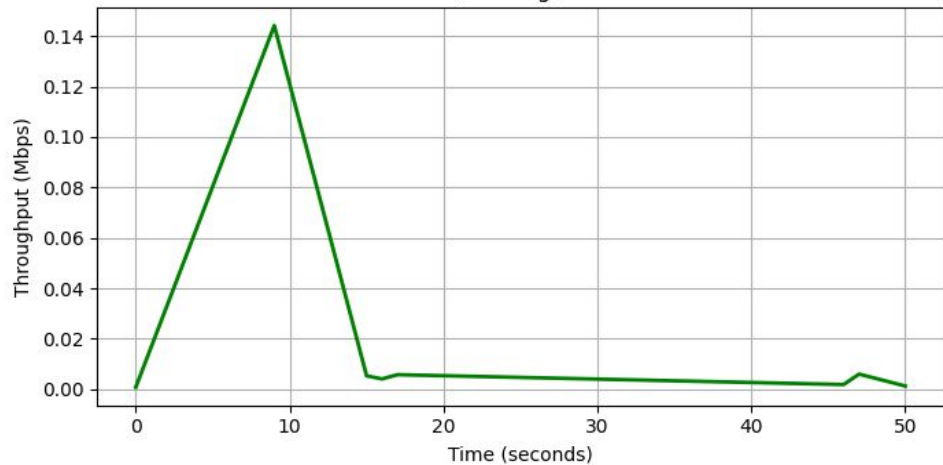
Zoom Bandwidth



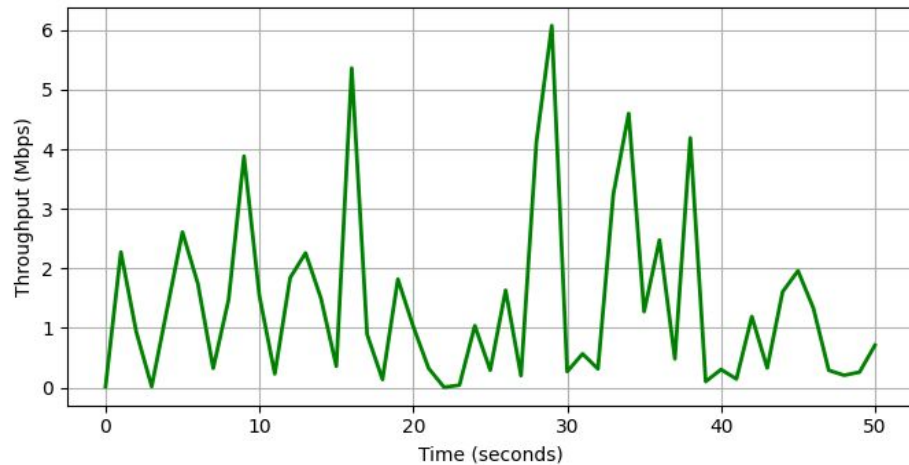
YouTube (Steady) Bandwidth



YouTube (Clicking) Bandwidth



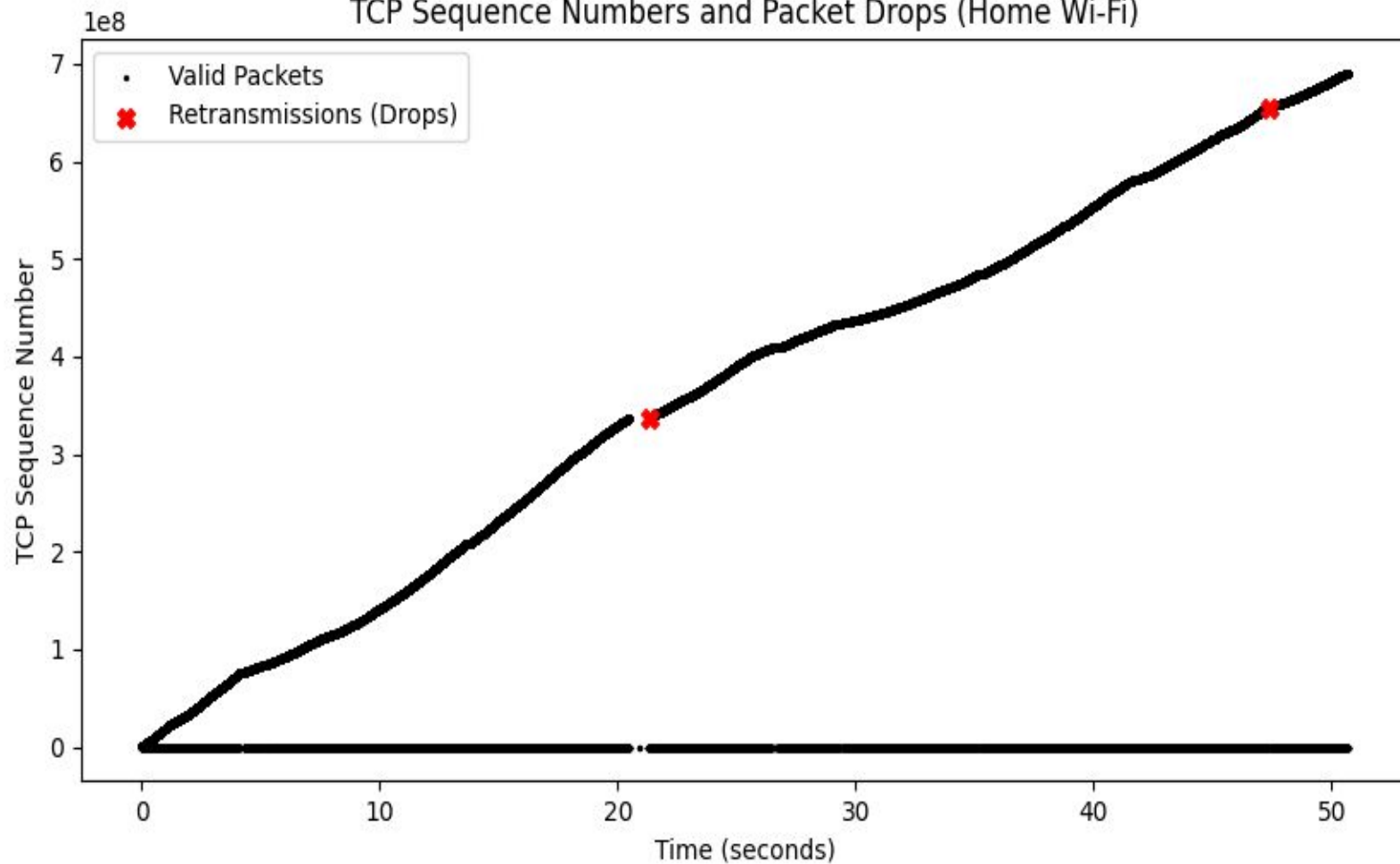
BBC News Bandwidth



TCP Packet Drops

- Linear Growth: The sequence number trajectory is nearly perfectly straight, showing uninterrupted data flow.
- Minimal Drops: Detected only two packet retransmissions across the entire 50-second test window.
- Routing Efficiency: Demonstrates that residential ISP routing experiences far less congestion than multi-hop enterprise routing on a campus.

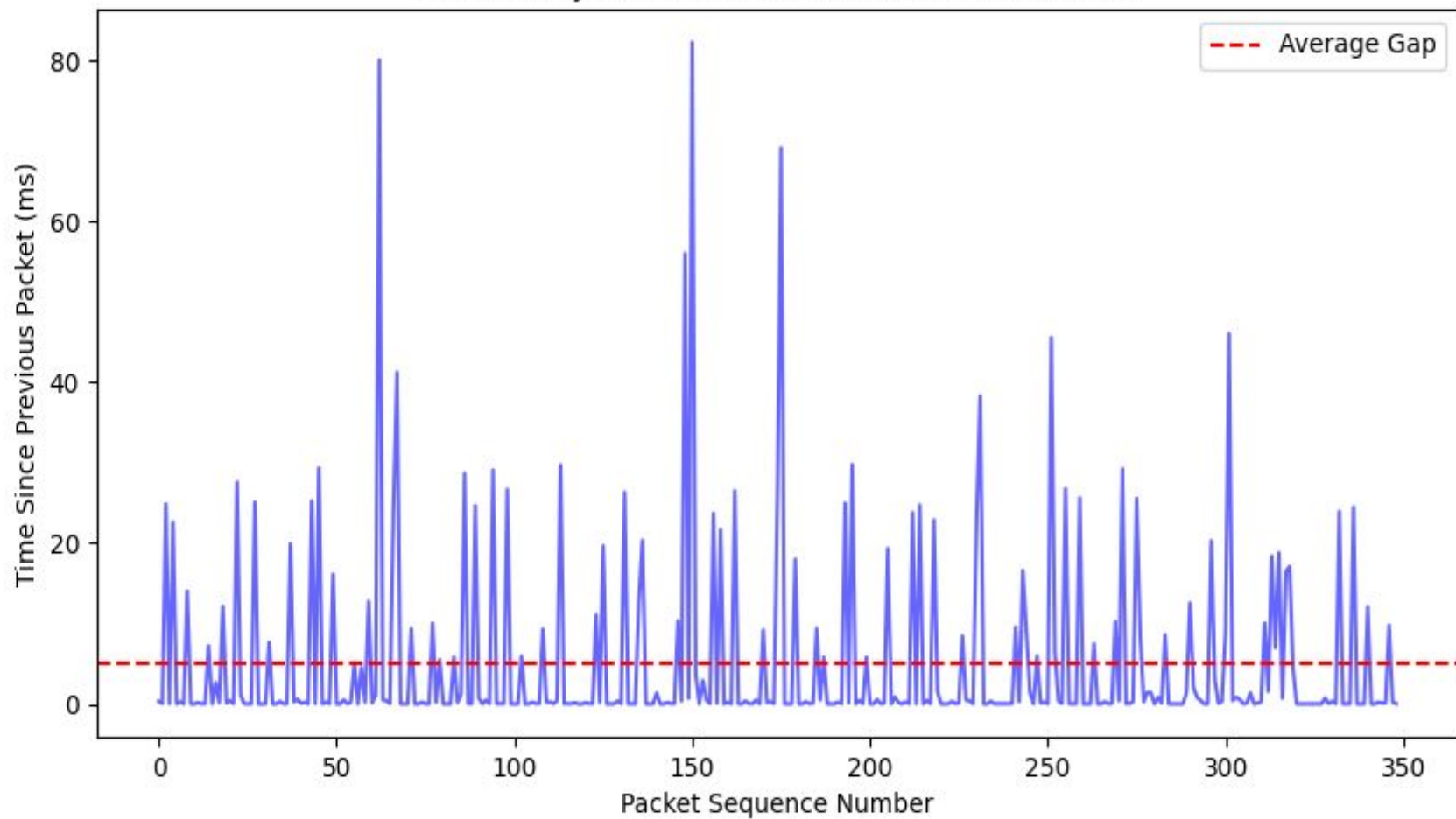
TCP Sequence Numbers and Packet Drops (Home Wi-Fi)



UDP Jitter

- Reduced Spikes: Jitter is strictly capped at 20-40ms, a significant improvement over the 100ms+ campus spikes.
- Tighter Gaps: Predictable packet arrival timing maintains smooth, synchronized media playback.
- Physical Layer Logic: With fewer nearby Access Points (APs), the device spends less time in MAC-layer “Backoff” mode and more time actively transmitting.

Zoom UDP Jitter on Home Wi-Fi (Low Interference)



Final Comparison and Conclusion

- 802.11 Contention: Wi-Fi interference on campus is the primary driver of UDP jitter, which directly degrades the real-time user experience.
- Enterprise Congestion: Multi-hop routing increases the probability of TCP packet drops, triggering retransmissions that throttle bulky file transfers.
- Application Resilience: Modern platforms successfully hide minor network failures, but their performance ceilings are ultimately bound by the physical Wi-Fi environment.
- Packet-Level Mastery: Analyzing traffic at the packet level successfully traces user-facing lag back to root causes like MAC layer backoff and TCP congestion windows.

Network Security & Protocol Workarounds

- The “Greedy” Nature of UDP: Because UDP lacks intrinsic congestion control, high-volume traffic can overwhelm routers, forcing apps onto polite TCP protocols when networks are busy.
- Infrastructure Strain: Tracking connectionless UDP traffic eats massive amounts of firewall memory, as there are no official handshake or teardown sequences.
- Security Vulnerabilities: UDP is frequently spoofed for DDoS attacks, prompting campus security to block non-standard ports.
- The Zoom Fallback: When strict firewall rules block its preferred UDP path, Zoom seamlessly disguises its traffic as standard web data over TCP Port 443 to keep the call alive.